

Technical Report N° 04 — Bridging Phase 1 and Phase 2

Project: *Forecasting Bitcoin Prices and Analyzing Geopolitical Risks: A Residual Analysis Approach using LSTM (Geo-BTC).*

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Phase: Inference & Causal Analysis (transition from Phase 1 → Phase 2).

Status: Closed.

Companion artefacts: `results/inference_table.csv`, `results/explaining_ratio.csv`, `reports/figures/inference_storyboard_2015.html`.

1 · Purpose

This report consolidates the results of the **quantitative phase** (LSTM forecasting + residual analysis) and introduces the **qualitative phase** (geopolitical and macroeconomic event database). It explains the feature-engineering performed by the merge step, documents the event categorisation that emerged from the qualitative research, and concludes with the rationale behind the visual flag system used in the inference storyboard.

2 · Phase-1 quantitative results (recap)

A univariate LSTM was trained on daily BTC/USD closing prices from the Bitstamp exchange, with the temporal split detailed in [docs/METHODOLOGY.md](#):

Subset	Period	Days
Train	2012-01-01 → 2014-09-30	1 005
Test	2014-10-01 → 2014-12-31	92
Validation (out-of-sample)	2015-01-01 → 2015-12-31	362

After hyper-parameter grid search the champion configuration (`look_back=90`, `units=100`) was retrained on the merged 2012-2014 block and used to perform out-of-sample inference for

every trading day in 2015.

For each inference day `t` the pipeline emits a residual record with columns

Date, Actual_Price, Predicted_Price, Abs_Error, Pct_Error, Directional_Error, Is_Outlier, Outlier_Type

Days outside the $\pm 2\sigma$ band on directional residuals are flagged as outliers and exported, with a $[-3, +7]$ -day asymmetric window, to `investigation\_periods.csv` — the input file consumed by Phase 2.

Sample of investigation_periods.csv

Event_Date	Actual_Price_USD	Error_Value_USD	Type	Start_Date	End_Date
2015-03-20	245.00	35.50	Drop	2015-03-17	2015-03-27
2015-08-19	215.00	90.20	Drop	2015-08-16	2015-08-26
2015-11-04	470.00	105.10	Peak	2015-11-01	2015-11-11
2015-12-10	440.00	42.80	Peak	2015-12-07	2015-12-17

3 · Phase-2 qualitative database

A parallel research effort produced

`data/external/qualitative\_analysis.csv` (also distributed as `qualitative\_analysis.xlsx`). Each row documents a 2015 event window with the schema

Date, Fluctuation, Category, Event (Macro), Breakdown (Micro), Price Movement

Date is a free-form **bilingual period string** — for example

`"15-25 Mar 2015", "23 Nov - 03 Dec 2015"` or

`"15-25 Ago 2015"`. The strings preserve the granularity of the sources (CoinDesk BPI, CNN Markets, BBC News, WSJ, Bitcoinity), but are not directly joinable with the LSTM exports.

Sample of the qualitative dataframe

Date	Fluctuation	Category	Event (Macro)
15-25 Mar 2015	Down	Ecosystem Vulnerability	Hot-wallet hacks reignite Mt. Go
15-25 Aug 2015	Down	Global Macroeconomic Shock	China devalues the yuan three ti
27 Oct - 06 Nov 2015	Up	Chinese Market Demand Surge	Volume on Chinese exchanges expl
30 Oct - 09 Nov 2015	Up	Global Expansion	Move stops being only Chinese, g

Eight categories emerge from the research, partitioned across two behavioural regimes:

Regime	Category
Down	Ecosystem Vulnerability · Global Macroeconomic Shock
Up	Institutional Interest · Chinese Market Demand Surge · Global Expansion · Technical Consolidation · Speculative Re-entry · Parabolic Move — FOMO

4 · Feature engineering of the merge

The bridge between phases is implemented by

``src/inference/merge.py``. Two design decisions are central:

4.1 · A bilingual period parser

The parser recognises three regular-expression patterns and a bilingual month dictionary (English + Portuguese):

Pattern	Example	Result
<code>`-`</code>	<code>`"15-25 Mar 2015"`</code>	(2015-03-15, 2015-03-25)
<code>`- -`</code>	<code>`"23 Nov - 03 Dec 2015"`</code>	(2015-11-23, 2015-12-03)
<code>`- + default_year`</code>	<code>`"05-15 Dec"`</code>	(default_year-12-05, default_year-12-15)

Range separators ``-``, ``-``, ``-`` and ``"to"`` are all accepted; the match is case-insensitive and tolerant to extra whitespace.

4.2 · Containment-based join with midpoint tie-breaking

Each LSTM outlier ``Event_Date`` is joined to the **single** qualitative window that contains it^{**}. When more than one window applies, the one whose midpoint is closest to the outlier wins. Outliers that fall outside every window keep ``Category = NaN`` — they remain in the inference table but contribute to the *un-explained* share of the Explaining Ratio.

The output table has the columns required by the meeting brief:

```
Event_Date, Data_Window, Real_Price, Absolute_Error,
Fluctuation, Category, Event_Macro, Breakdown_Micro, Price_Movement
```

5 · Category split — Phase 2 contribution to the analysis

The `summarise_by_category` helper aggregates the matched outliers by Category, computing per-category MAE and the share of total identified outliers — the **per-category Explaining Ratio**. Summing the ER column reproduces the global ratio.

A representative slice (numerical values depend on the active grid search and threshold; see ``results/explaining_ratio.csv`` for the canonical numbers):

Event Category	Frequency	MAE (USD)	ER (% of total)
Global Macroeconomic Shock	...	...	...
Chinese Market Demand Surge	...	...	...
Ecosystem Vulnerability	...	...	...
...	...	...	...

6 · The visual flags — and the report decision

The inference storyboard

(``reports/figures/inference_storyboard_2015.html``) overlays the **Real vs Predicted** time series and annotates only those outliers whose absolute error exceeds $\theta = 20 \text{ USD}$ — the *Rule of 20* adopted in Phase 2. Each flag carries:

- A red triangle marker at the actual-price point of the outlier day.
- An anchored annotation with the truncated `Event_Macro` text.
- A hover tooltip with the full event description, category and absolute error.

Two reasons drove the choice of an **interactive HTML** storyboard:

1. The reader needs to inspect *individual* annotations without losing the context of the full year — Plotly's hover and zoom satisfy that requirement; a static PNG would not.
2. The number of annotations (typically 10-15) makes a fully static layout unreadable.

A static PNG companion is generated automatically when the optional ``kaledgeo`` dependency is present; its sole purpose is to embed the

figure inside the manuscript.

7 · Conclusion

Phase 2 closes the loop opened by Phase 1:

Phase 1 told us **where** the market behaved unpredictably.

Phase 2 tells us **why**.

The merge engineering documented above formalises this bridge and exports two reproducible artefacts (`inference_table.csv` and `explaining_ratio.csv`) and one interactive figure (`inference_storyboard_2015.html`). All three feed directly into the results section of the paper.

The next deliverable, **Technical Report N° 05**, documents the research process behind the qualitative database and presents the flag-level overview that complements the aggregate metrics reported here.